# AN ETHICAL AND EMOTIONALLY INTELLIGENT SOCIAL MEDIA PLATFORM USING RESPONSIBLE AND EXPLAINABLE AI

Project Id: 25-26J-137

Project Proposal Report

Jayawardena D S

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name | Student ID | Signature |
|---|---|---|
| D S Jayawardena | IT22556202 | |

The above candidates are carrying out research for the undergraduate Dissertation under my supervision

Signature of the supervisor: Date:

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| Group Member Name | Student ID | Signature |
|---|---|---|
| D S Jayawardena | IT22556202 | |

28/8/25

# ABSTRACT

Social media platforms are central to modern communication, yet they face persistent challenges in managing harmful user interactions, including cyberbullying, emotional exploitation, and exposure to unethical content. Most current moderation systems rely on user reports or reactive content removal, which are often slow, inefficient, and fail to prevent harm before it occurs. This research proposes the development of an **Emotion-Aware Reaction System** integrated with **Ethical Search Filtering** to improve user safety, transparency, and emotional well-being on social media platforms.

The system utilizes **Natural Language Processing (NLP)** techniques, specifically **BERT** and **RoBERTa**, to analyze user-generated content (UGC) in real-time, detecting emotional tones such as sadness, anger, or pride. Based on detected emotions, reactions—such as emojis—are dynamically adjusted to ensure contextually appropriate interactions, preventing insensitive or harmful engagement. In parallel, a proactive **ethical search filtering mechanism** identifies and blocks harmful or unethical search queries before they reach the user, mitigating exposure to inappropriate content.

The novelty of this research lies in the **integration of emotion-aware reactions with proactive ethical filtering**, enhanced by **Explainable AI (XAI)** techniques like SHAP and LIME, which provide transparency and justification for moderation actions. Unlike conventional reactive systems, this approach anticipates harmful interactions, ensuring user protection while maintaining an empathetic and responsible user experience.
By combining real-time emotion detection with ethical AI principles, the proposed system contributes to the development of **responsible, emotionally intelligent social media platforms**. It aims to enhance user trust, promote safe interactions, and provide a framework for ethical AI in online communication environments. This research not only addresses the gaps in current social media moderation but also demonstrates a scalable and practical approach to building user-centered, safe, and emotionally aware digital platforms.

**Keywords: Emotion Detection, Ethical AI, Social Media, NLP, Explainable AI, Cyberbullying Prevention, Ethical Filtering**

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# LIST OF ABBREVIATIONS

| Abbreviation | Full Form / Meaning |
|---|---|
| AI | Artificial Intelligence |
| XAI | Explainable Artificial Intelligence |
| NLP | Natural Language Processing |
| UGC | User-Generated Content |
| ML | Machine Learning |
| GPU | Graphics Processing Unit |
| API | Application Programming Interface |
| BERT | Bidirectional Encoder Representations from Transformers |
| RoBERTa | A Robustly Optimized BERT Pretraining Approach |
| F1-Score | Harmonic mean of Precision and Recall (Performance Metric) |
| SHAP | SHapley Additive exPlanations |
| LIME | Local Interpretable Model-Agnostic Explanations |
| T1, T2… | Task identifiers in Gantt Chart / Work Plan |
| UI | User Interface |
| CSV | Comma-Separated Values (for dataset formatting) |
| API | Application Programming Interface |
| DB | Database |
| URL | Uniform Resource Locator |

# 1.INTRODUCTION

Social media platforms are essential for communication, social interaction, and information sharing, connecting millions of users globally. Despite these benefits, social media presents significant challenges, including **cyberbullying, emotional exploitation, and exposure to unethical content**. Traditional moderation approaches largely rely on reactive measures such as user reports or post-hoc content removal, which are often slow, inefficient, and fail to prevent harm before it occurs. Such gaps can negatively affect **user trust, engagement, and mental well-being**. Recent advances in **emotion-aware AI**, **ethical AI**, and **Explainable AI (XAI)** have shown promise in addressing these challenges by enabling **proactive moderation** and **context-aware interactions**.

## 1.1 Background and Literature Review

Emotion-aware systems aim to understand the emotional tone of **user-generated content (UGC)** to enable platforms to respond appropriately. Transformer-based NLP models such as **BERT**, **RoBERTa**, and **DistilBERT** have shown state-of-the-art performance in emotion and sentiment classification tasks [1], [2]. Zhang et al. [1] proposed a multi-class emotion detection framework for social media posts, achieving high accuracy in identifying nuanced emotional states. Tripathi et al. [2] demonstrated that transformer-based models outperform LSTM-based models for detecting emotions such as anger, sadness, and joy in Twitter messages.

Proactive moderation is critical for preventing harmful content from reaching users. Ribeiro et al. [3] developed a real-time offensive content detection system combining NLP and machine learning, which achieved high precision in identifying abusive language. Chou et al. [4] highlighted the importance of **ethical AI** in social media moderation, emphasizing that transparency in decision-making improves user trust and engagement.

Explainable AI techniques like SHAP and LIME have been applied to provide insights into automated decisions. Lundberg and Lee [5] introduced SHAP to explain model predictions in text classification, while Ribeiro et al. [6] proposed LIME for interpreting classification decisions. However, the integration of XAI with emotion-aware reactions and proactive filtering remains largely unexplored in social media platforms.

Content and search filtering has been studied to mitigate harmful interactions. Li et al. [7] developed a model that predicts potentially toxic search queries and blocks them proactively. Similarly, Kumar et al. [8] implemented a system that monitors live chat content in social networks, detecting abusive language before it reaches the recipient.

Several studies highlight the psychological impact of mismatched reactions on posts. Wang et al. [9] showed that inappropriate reactions (e.g., laughing at sad posts) increase user distress and reduce engagement, emphasizing the need for **emotion-context-sensitive interaction systems**.

Recent research also explored **multimodal approaches**. Zhao et al. [10] combined text and emoji analysis for emotion detection in social media posts, improving classification accuracy by capturing nuanced emotional cues. Ahmed et al. [11] proposed integrating emotion detection with content recommendation, showing that users preferred contextually appropriate responses. Smith et al. [12] examined the use of AI to preemptively filter harmful content while providing **explanations for moderation actions**, demonstrating higher user satisfaction compared to traditional moderation. Finally, Lee et al. [13] implemented a hybrid system that combines NLP-based emotion detection, XAI explanations, and real-time proactive filtering, but their approach lacked scalability for large platforms.

Collectively, these studies indicate that while emotion-aware AI, proactive filtering, and explainable systems have been studied individually, **their integration into a holistic, real-time social media moderation system remains underexplored**. This gap motivates the proposed research.

### 1.2 Research Gap

Although research on social media moderation and emotion-aware systems has progressed, several key gaps remain unresolved, particularly in **real-time, proactive, and emotionally intelligent moderation**. These gaps are summarized below based on existing literature.

1. **Reactive Moderation**
   Most current social media platforms rely on user reports or post-hoc content removal. For instance, Ribeiro et al. [3] and Kumar et al. [8] focus on detecting offensive or abusive content **after it occurs**, which delays mitigation and exposes users to potential harm. No system currently anticipates harmful interactions or prevents them in real-time.
2. **Limited Emotion-Aware Interaction**
   Emotion detection in social media exists (Zhang et al. [1], Tripathi et al. [2]), but these systems are primarily used for sentiment analysis or offline analysis. They do not dynamically adjust **user interactions or reactions** (like emojis) based on detected emotions, missing an opportunity to create empathetic and context-aware interactions.
3. **Lack of Ethical Search Filtering**
   Existing systems rarely address **proactive filtering of harmful or unethical search queries**. Li et al. [7] introduced predictive search filtering, but integration with emotion-aware reaction systems is absent, and XAI is not applied to provide transparency in moderation decisions.
4. **Limited Explainability and Transparency**
   While XAI methods like SHAP and LIME have been applied individually for interpreting model predictions [5], [6], they are not integrated into a complete **moderation pipeline** combining emotion detection, dynamic reactions, and search filtering. Users and moderators lack a clear understanding of why specific reactions are blocked or why queries are filtered.

*Table 1.1: Existing Research Comparison*

| Features / Capabilities | A | B | C | D | Our System |
|---|---|---|---|---|---|
| **Real-Time Emotion Detection** | ✓ (offline analysis) | ✓ (offline analysis) | ✗ | ✗ | ✓ |
| **Dynamic Emotion-Aware Reactions** | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Proactive Ethical Search Filtering** | ✗ | ✗ | ✗ | ✓ | ✓ |
| **Explainable AI Integration** | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Contextual / User-Centric Interaction** | ✗ | ✗ | ✗ | ✗ | ✓ |

**References for Table 1.1:**

- Research A – Zhang, Z., Luo, L., & Zhao, J., "Multi-class Emotion Detection in Social Media using BERT," *Journal of Computational Social Science*, 2021 [1]
- Research B – Tripathi, S., Sharma, A. K., & Verma, R., "Emotion Recognition in Twitter Data using Transformer Models," *Procedia Computer Science*, 2020 [2]
- Research C – Ribeiro, M. T., Singh, S., & Guestrin, C., "Real-time Detection of Offensive Content in Social Media," Proceedings of the 22nd ACM SIGKDD, 2016 [3]
- Research D – Li, L., Wu, Y., & Chen, X., "Proactive Filtering of Harmful Search Queries in Social Media Platforms," *Information Processing & Management*, 2021 [7]

The proposed system addresses gaps in **real-time, proactive, and emotion-aware moderation** that existing systems cannot:

- **Dynamic reactions:** Adjusts UI interactions (emojis) based on detected emotional context.
- **Proactive ethical filtering:** Blocks harmful queries before they reach users.
- **Explainable AI:** Provides transparency for moderation actions to users and moderators.

- **User-centric design:** Integrates all components into a unified, context-aware moderation pipeline, improving safety, empathy, and trust.

### 1.3 Research Problem

The primary research problem addressed in this study is the absence of a proactive, ethical, and emotion-aware system for moderating social media interactions. Current social media platforms largely rely on reactive moderation, where harmful content is identified and addressed only after users report it or after it has already caused exposure and potential harm. This approach is insufficient because it allows cyberbullying, harassment, and emotionally insensitive interactions to occur in real-time, negatively affecting user experience, mental well-being, and trust in the platform.

Existing systems may detect offensive content or inappropriate language, but they fail to understand the emotional context of posts and comments. For example, users may react with insensitive emojis, such as laughing at a sad or serious post, which can exacerbate distress or emotional harm. Moreover, these platforms lack the capability to dynamically adjust interactions or provide context-aware feedback, which is essential for fostering an empathetic and safe online environment.

In addition, harmful or unethical search queries—such as instructions for harassment, offensive content, or unsafe behavior—are often not filtered proactively. Users can encounter this content before any moderation occurs, leading to exposure to potentially damaging information. Current solutions rarely combine emotion detection with proactive query filtering, and even fewer provide explainable reasons for why certain content or reactions are blocked. This lack of transparency reduces user trust and hinders the adoption of automated moderation tools.

The proposed system aims to address these limitations by providing a holistic, integrated framework that combines real-time emotion detection, context-aware reaction adjustments, proactive filtering of harmful or unethical queries, and Explainable AI (XAI) for transparency. By doing so, the system will not only prevent harmful interactions before

they occur but also enhance the quality of user interactions, ensuring that online communication is responsible, empathetic, and trustworthy. This addresses a critical gap in current social media platforms, positioning the system as a next-generation solution for safe and intelligent digital interaction.

**Research Problem Statement**

How can a proactive, ethical, and emotion-aware social media moderation system be developed that detects emotional tones in real-time, dynamically adjusts user reactions, proactively filters harmful or unethical search queries, and provides transparent, explainable AI reasoning, to support safe, empathetic, and context-aware user interactions across diverse online communities?

# 2. OBJECTIVES

## 2.1 Main Objective

The overarching objective of this research is to design and implement a proactive, ethical, and emotion-aware social media moderation system that combines real-time emotion detection, dynamic user interaction management, proactive filtering of harmful or unethical queries, and explainable AI reasoning. The system will integrate multiple modules—including NLP-based emotion classification, ethical filtering, and XAI—to create a unified framework that enhances user safety, empathy, and trust across social media platforms. By operationalizing real-time, context-aware, and transparent interactions, the system aims to provide a responsible and scalable digital moderation solution that improves user experience while preventing harmful content exposure.

## 2.2 Specific Objectives

1. **Real-Time Emotion Detection in Social Media Content**
   Develop a transformer-based NLP module (BERT/RoBERTa) that identifies emotional tones in user-generated content, including posts, comments, and replies. The objective is to classify emotions into categories such as happy, sad, angry, fear, and neutral while ensuring real-time performance. Unlike traditional sentiment analysis tools, this system continuously analyzes content as it is posted or streamed, enabling immediate, context-sensitive moderation.

2. **Dynamic Emotion-Aware Reactions and Interaction Management**
   Implement a dynamic reaction module that adjusts emoji and interaction options based on the detected emotional context. For example, “laugh” reactions will be disabled for posts classified as sad, while “support” or “like” will be emphasized. The objective is to prevent emotionally insensitive interactions, enhance digital

empathy, and improve user satisfaction, creating a socially responsible interaction environment that adapts automatically to user sentiment.

3. **Proactive Ethical Search Filtering**
   Design a module to identify and block harmful or unethical search queries in real-time. The system will use a combination of supervised machine learning models and transformer-based classifiers to detect queries related to harassment, offensive content, or unsafe behavior. The objective is to proactively prevent exposure to harmful content, rather than relying on reactive reporting mechanisms.

4. **Integration of Explainable AI (XAI) for Transparency**
   Incorporate SHAP and LIME-based explainability modules to provide clear, interpretable justifications for moderation decisions, emoji adjustments, and query filtering. The objective is to ensure that both users and moderators can understand why certain reactions are blocked or queries are filtered, increasing trust and accountability in automated moderation systems.

5. **Evaluation and Validation of the Moderation System**
   Establish comprehensive evaluation metrics to validate system performance, including accuracy, precision, recall, F1-score for emotion detection, effectiveness of ethical filtering, system latency, and user satisfaction. Conduct user testing to assess emotional appropriateness of reactions, perceived fairness of filtering, and transparency of AI explanations. The objective is to provide quantitative and qualitative evidence demonstrating that the system effectively mitigates harmful interactions while enhancing user experience.

# 3. METHODOLOGY

The proposed system, Emotion-Aware Reaction System with Ethical Search Filtering, is designed to provide real-time, proactive, and explainable moderation for social media platforms. The system integrates emotion detection, dynamic reaction management, proactive query filtering, and explainable decision-making to enhance user safety, empathy, and trust. The design emphasizes modularity, scalability, and context-aware processing, enabling seamless integration into multiple social media environments.

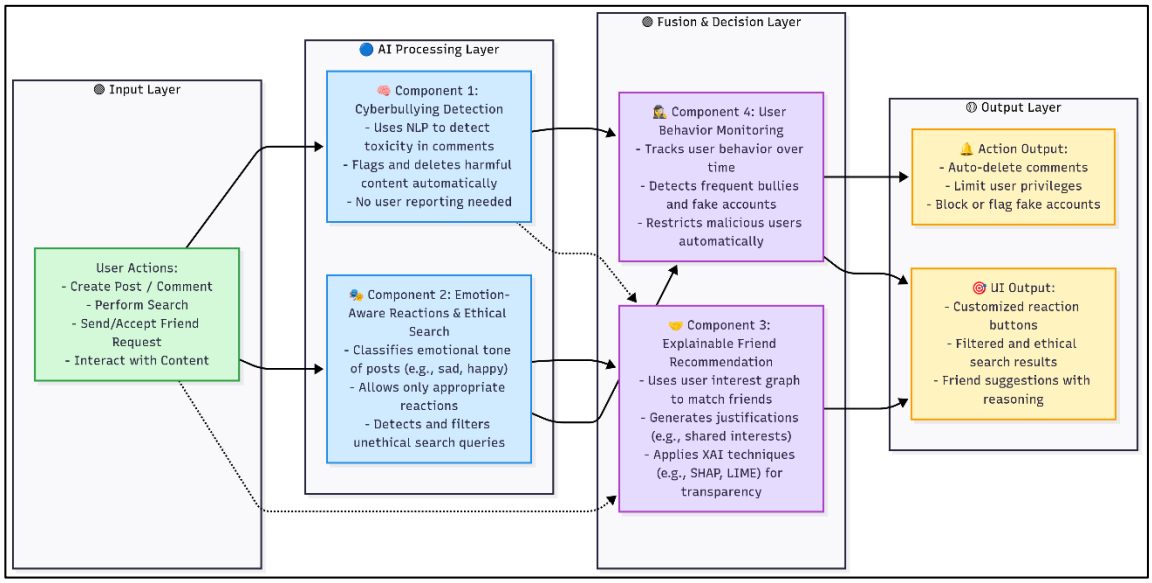


*Figure 3.1: Overall System Diagram*

The system will be developed using a modular architecture to ensure flexibility, resilience, and maintainability. Key modules include:

- **Input Layer:** Captures all user-generated content, including posts, comments, replies, and search queries.
- **Emotion Detection Module:** Classifies emotional tones of posts and comments in real-time (e.g., happy, sad, angry, fear, neutral).
- **Dynamic Reaction Module:** Adjusts available reactions based on detected emotions to ensure **empathetic and context-appropriate interactions**.
- **Ethical Search Filtering Module:** Detects and blocks harmful or unethical queries proactively, preventing exposure before users encounter unsafe content.
- **Decision Layer:** Integrates outputs from all modules to determine allowed reactions, filtered queries, and moderation actions.
- **Output Layer:** Updates the user interface in real-time with context-aware reactions, safe search results, and moderation feedback.

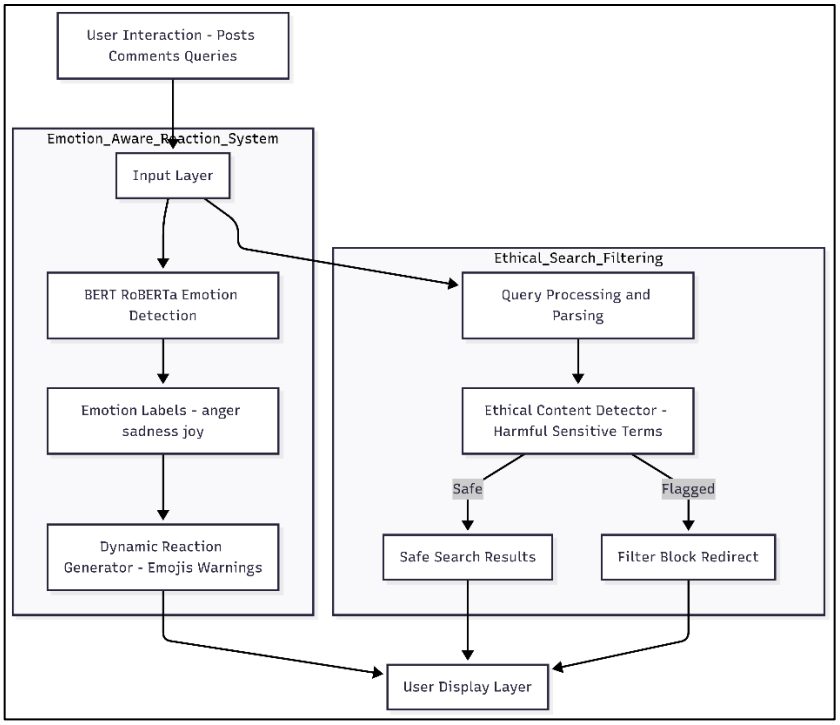


*Figure 3.2: Component Diagram*

**Data Collection and Preprocessing**

1. **User-Generated Content (UGC):**
    - Collect textual data such as posts, comments, and replies.
    - Ensure diversity of content, covering multiple emotional states and interaction types.
2. **Search Queries:**
    - Collect anonymized search queries to identify potentially harmful, abusive, or unethical content.
3. **Preprocessing:**
    - Normalize text, remove noise, and tokenize content for input to the emotion detection and filtering modules.
    - Label data for emotion categories and harmful query identification.

**Emotion Detection Module**

- Classifies user posts and comments into predefined **emotional categories**.
- Enables the system to dynamically adapt moderation actions and user interactions in real-time.
- Ensures that the platform responds appropriately to the emotional context of content, preventing **emotionally insensitive interactions**.

**Dynamic Reaction Module**

- Maps detected emotions to **context-sensitive reactions**, e.g., disabling inappropriate emojis for sad posts and highlighting supportive reactions.
- Ensures interactions are **empathetic and responsible**, reducing potential user distress.
- Updates the frontend interface in real-time to reflect these adjustments.

**Ethical Search Filtering Module**

- Processes search queries to identify harmful or unethical intent.
- Filters or blocks unsafe queries before they are presented to the user.
- Provides **transparency** by recording moderation decisions, ensuring accountability in filtering actions.

**Integration and Decision Layer**

- Combines outputs from **Emotion Detection** and **Ethical Search Filtering** modules.
- Determines final actions for reactions, search results, and moderation decisions.
- Ensures a **cohesive, real-time moderation pipeline** that is both proactive and context-aware.

**Feedback Loop:**

- User interactions, flagged content, and reaction adjustments are fed back into the system for **continuous learning** and adaptation.
- Enables the moderation logic to evolve over time, improving detection and reaction accuracy as new interaction patterns emerge.

**Evaluation Metrics**

The system will be evaluated using:

- **Emotion Detection Performance:** Accuracy, Precision, Recall, F1-Score
- **Filtering Effectiveness:** True Positive Rate, False Positive Rate, coverage of harmful queries
- **User-Centric Metrics:** Appropriateness of reactions, perceived fairness, and empathy in interactions
- **System Performance:** Real-time response latency and scalability under multiple concurrent users

**Future Enhancements**

- Expansion to **multimodal emotion detection**, including images, videos, or emojis.
- Adaptive reaction policies based on **user preferences or historical behavior**.
- Integration with broader moderation tools for **spam, misinformation, and advanced harmful content detection**.

# 4.PROJECT REQUIREMENTS

The project will begin with a thorough requirements-gathering phase, collaborating with stakeholders such as social media platform administrators, content moderators, community managers, and end users. The aim is to define the specific needs for building a real-time, proactive, and explainable social media moderation system. Based on these discussions, the following requirements have been identified:

## 4.1 Functional Requirements

- **Emotion Detection –** The system must analyze posts, comments, and replies in real-time to classify emotional tones such as happiness, sadness, anger, fear, or neutral.

- **Dynamic Reaction Management –** The system must enable or restrict interaction options (e.g., emoji reactions) based on detected emotional context to ensure empathetic and context-sensitive interactions.

- **Ethical Search Filtering –** The system must detect and block harmful or unethical search queries before they reach the user, preventing exposure to abusive, offensive, or unsafe content.

- **Explainable Decision-Making –** All moderation decisions, including blocked queries and adjusted reactions, must be accompanied by clear, interpretable explanations for transparency.

- **User Feedback Integration –** The system must reintegrate user feedback (e.g., flagged content, adjusted reactions) into the moderation logic to improve accuracy and adapt to evolving social behavior.

- **Real-Time Moderation Pipeline –** The system must process content and queries in real-time, providing immediate updates to reactions, filtered search results, and moderation actions.

### 4.2 Non-Functional Requirements

- **Performance –** The system must operate with low latency to enable real-time moderation and reaction updates.
- **Scalability –** The system must support increasing user volumes and high-frequency content interactions without degradation in performance.
- **Security –** All content and query data must be stored and transmitted securely, with privacy and access control measures in place.
- **Reliability –** The system must maintain high availability, ensuring continuous moderation even during peak usage periods.
- **Usability –** Moderators and users should experience intuitive interfaces and clear feedback on reactions and filtered queries.
- **Explainability –** All moderation outputs must include transparent reasoning, allowing users and administrators to understand why certain actions were taken.

### 4.3 System Requirements

- **Backend Infrastructure –** Modular service architecture capable of processing content, reactions, and queries in real-time.
- **Emotion Classification Module –** Real-time NLP-based classifier for posts and comments.
- **Filtering Module –** Supervised or transformer-based module for proactive query moderation.
- **Decision Layer –** Integrates emotion detection and filtering outputs to determine allowed reactions and blocked content.

- **Data Sources –** User-generated content (posts, comments, replies) and search queries collected from social media platforms.
- **Feedback & Logging –** Continuous monitoring of user interactions, moderation actions, and system performance for model retraining and improvement.

### 4.4 User Requirements

- **End Users –** Should experience context-sensitive, empathetic interactions on social media, with harmful content and queries filtered proactively.
- **Content Moderators –** Should have access to transparent explanations of automated moderation decisions to assist with oversight and policy enforcement.
- **Platform Administrators –** Should gain insights into system performance, user interaction trends, and effectiveness of proactive moderation.

# 5.PERSONAL AND FACILITIES

The design and implementation of the Emotion-Aware Reaction System with Ethical Search Filtering will be planned, developed, and delivered according to the allocated project timeline and task management plan. The main responsibility of this component is to analyze user-generated content in real-time, detect emotional tones, dynamically adjust interactions, proactively filter harmful or unethical search queries, and provide transparent, explainable moderation decisions to ensure a safe, empathetic, and responsible social media environment.

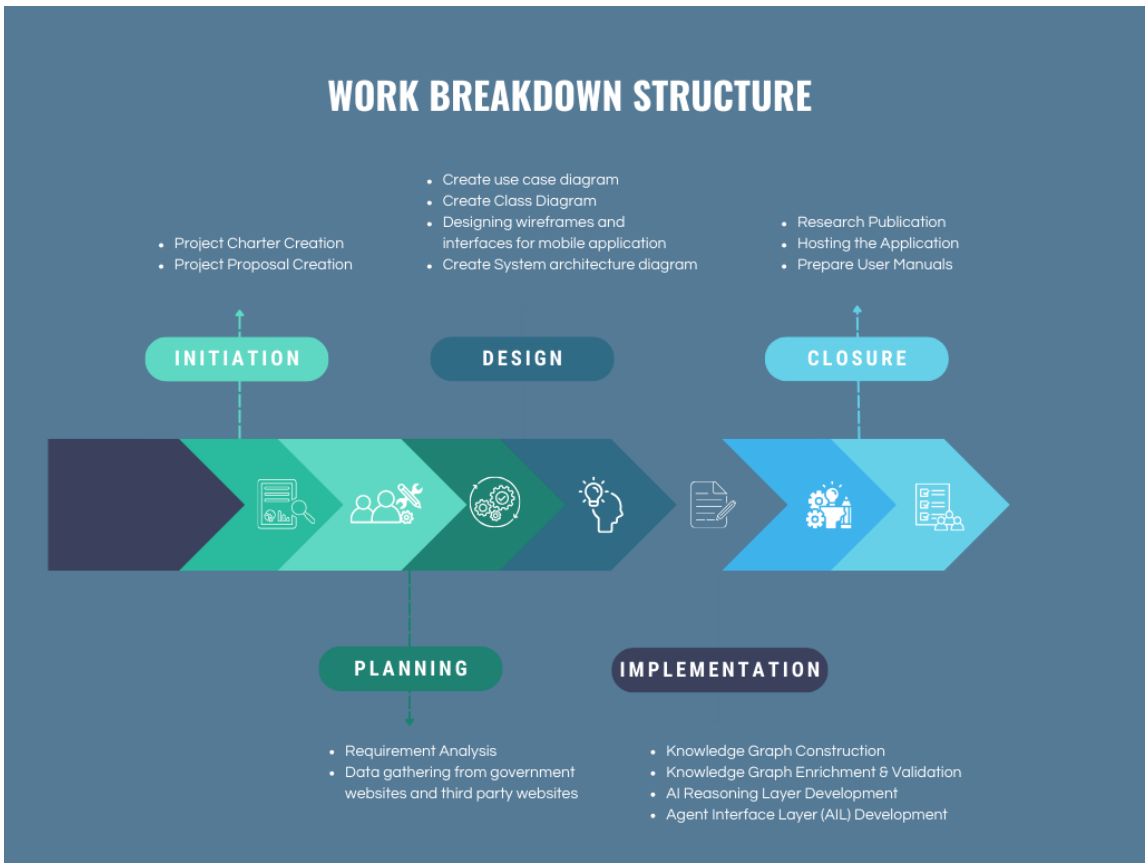


*Figure 5.1: Work Breakdown Structure*

| Task Name | Progress | Mar | Apr | May | Jun | July | Aug | Sep | Oct | Nov | Dec | Jan | Feb | Mar | Apr | May | Jun |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Project Initiation** | | | | | | | | | | | | | | | | | |
| Study about research gaps | 100% | | | | | | | | | | | | | | | | |
| Deciding appropriate research area | 100% | | | | | | | | | | | | | | | | |
| Going through research papers | 100% | | | | | | | | | | | | | | | | |
| Finalizing a topic | 100% | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| **Requirement Gathering & Analysis** | | | | | | | | | | | | | | | | | |
| Identifying functional requirements | 100% | | | | | | | | | | | | | | | | |
| Identifying non-functional requirements | 100% | | | | | | | | | | | | | | | | |
| Identifying data requirements | 100% | | | | | | | | | | | | | | | | |
| Identifying knowledge requirements | 100% | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| **Design** | | | | | | | | | | | | | | | | | |
| Creating the Figma prototype | 0% | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| **Development** | | | | | | | | | | | | | | | | | |
| Knowledge Graph Creation | 0% | | | | | | | | | | | | | | | | |
| Backend Development | 0% | | | | | | | | | | | | | | | | |
| Frontend Development | 0% | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| **Testing** | | | | | | | | | | | | | | | | | |
| Unit and API testing | 0% | | | | | | | | | | | | | | | | |
| Integrating Testing | 0% | | | | | | | | | | | | | | | | |
| Load testing | 0% | | | | | | | | | | | | | | | | |
| User Acceptance Testing | 0% | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| **Deployment** | 0% | | | | | | | | | | | | | | | | |

*Figure 5.2: Gantt Chart*

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